Journal of Chinese Human Resources Management
2023, VOL. 14, NO. 1, 38-59
DOI: 10.47297/wspchrmWSP2040-800502.20231401

A Systematic Bibliometric Analysis on the Current Digital Human Resources Management Studies and Directions for Future Research

Hale Alan

Akdeniz University, Manavgat, Antalya, Turkey

ABSTRACT

Electronic Human Resources Management (e-HRM) activities should be continued with more technological and innovative methods. The aim of this research is to examine in detail the current studies on e-HRM based activities and to determine the orientation of the concepts. As a result of keywords analysis, it is seen that big data, artificial intelligence, cloud computing, Internet of Things have come to the fore. In the last ten years, 3888 publications related to e-HRM have been retrieved from the web of science database, and 56 newest articles on Human Resource Management (HRM) application, practice, strategy and execution, which contain search phrases. It is seen that the subject of e-HRM is the starting point in determining the research terms and the main starting point of the articles that guide the study, and the most cited publications in this field are the studies on e-HRM. It is thought that developments in mobile applications, smart automation and artificial intelligence also affect the studies in the field of electronic or digital HRM. The common feature of the most recent publications on technology-oriented HRM practices, strategies and practices is the acceptance that their effects are wide-ranging. Recent discussions on technology and HRM strategies in businesses have emphasized the importance of social context and appear to have developed frameworks that acknowledge both the material and social character of technologies. As a result, the current increase in the use of digital technologies affects the competencies and expectations of the digital workforce.

KEYWORDS

E-HRM; Digitalization; HRM practices; HRM strategies; Mobile HRM.

1 Introduction

HRM is a process where processes involving many activities from recruitment to payroll management must be carried out simultaneously. When all the work in this process is carried out with traditional methods, more workload arises. When a most basic process activity is carried out with traditional methods, it can turn into a big problem for both the HR unit and the employees. This situation requires that this process be continued with more technological and innovative methods. Technology changes the way HR departments communicate with employees, store files and analyze employee performance. When used well, technology makes HR practices more efficient. In malicious use, it can get in the way of the company's human resources management. Good HR practices maximize benefits and minimize problems. These applications are electronic, digital, technology and informatics-based mobile or web applications. It can be said that these concepts have spread and developed under the umbrella of e-hrm. Although the most up-to-date is mobile application, it offers a more convenient and faster workflow than desktop or laptop computer (Shankar, 2022). As applications

Copyright: © 2023 Author(s). Published by World Scientific Publishing House. This is an open-access article under the CC BY 4.0 license (https://creativecommons.org/licenses/by/4.0/).

for processes such as recruitment and selection of personnel, wages, performance, rewards and incentives, and employee satisfaction increase the level of participation of employees in HR processes, it is easier for HR to determine and implement appropriate solutions and processes. In recent years, internet usage habits have changed rapidly and the number of people actively using social media is increasing day by day. The experience that social networks provide to people is reflected in many issues, especially in job application and recruitment processes. At this point, we can say that the application forms are gradually obsolete in the recruitment process. Even interviews, which are one of the important stages of the recruitment process, are now done through online platforms. Making the application process through software increases traceability and reportability.

Today, all countries in the world are inevitably affected by digital transformation. Mobile applications provide access to information from anywhere at any time of the day. For this reason, the ease of accessing and transacting job postings in mobile applications makes recruitment processes more enjoyable and practical. We see that not only the recruitment process but also career applications are gradually mobile. Advances in smart automation and artificial intelligence have necessitated human resources that are more qualified. Companies are now becoming mobile companies that can rapidly develop and market rapidly developing, flexible products. Many companies can look digital nowadays. However, very few businesses can be said to be digital. Businesses have learned to transform paper media into digital workflows. Using internal business portals, video chat, social media and many online tools, companies can go digital. However, being digital requires much more as a process that should involve all employees, focusing on solutions covering all departments and functions of the business.

An informatics and technology-oriented workforce, increasing with globalization, seeks optimum adaptation to maximize innovation (Caligiuri et al., 2020). Because of the challenges businesses face, they are expected to turn to creative workplace improvements to support their success. HRM plays an effective and important role in dealing with digital work, digital employee management, and digital work functions (Strohmeier & Parry, 2014). On the one hand, digital technologies continue to develop day by day, and on the other hand, new HRM applications that make it necessary to increase the quality and capacity of human resources have revealed that digital human resources management applications are an important issue to study and research. Digital technologies have a great contribution to redefining, redesigning and recombining human resources functions and interventions to create harmony with the success of achieving organizational goals, being productive in the focus of employee engagement and prevalence with digital human resource management practices. At the same time, personnel departments still face challenges in adopting new technologies, and e-HRM results are not always as positive as one might think. Digital technologies have an impact on the field of human resources practices in three ways. First, it is the workforce with practical skills to manage the digitalization process; the digital workplace, which includes the use of work environments and communication tools that will enable digital work to be done; and finally, the effects of digital HRM that will enable digital work to be done. Developing digital tools and equipment by developing solutions that match the right person with the right job necessitates changes in the field of HRM (Averyanov et al., 2017). Using e-HRM technology is a way to use HR strategies, policies and practices in the most efficient way.

In this study, a research based on the systematic bibliometric analysis method was carried out on the reflections of digital transformation on e-HRM. Within the scope of the research, human resources management practices, which have gained a new dimension with the effect of digital age and digitalization, and the concepts associated with digitalized human resources practices, the trends of current research on digital human resources practices and emerging new practices have been extensively researched and explained. The main research question to be answered is what are the features of the newest publications on technology-oriented developing HRM strategies and practices, and what the orientations of the focus of these publications are. At the same time, it is aimed to reveal the status of e-HRM studies and the directives of future studies by using different and comprehensive search terms.

2 Literature review

In the digital world, a new working environment provides an environment for new collaborative working models due to cloud computing platforms. Cloud computing, advanced algorithms, and artificial intelligence are interrelated and have a huge impact on how individuals act inside and outside the organization. Digitization has been shown to drastically change the workforce population, the skill set needed, and the ways in which they interact and collaborate. Within an organization, communication is carried out not only by the employee but also by the leadership. Businesses working on cloud technologies enable rapid development of business in a continuous and real-time manner. Digital HR technology has been defined as a set of software, hardware, and electronic network resource configurations designed and adopted to automate HR function in organizations using the latest information technology disruptions (Jani et al., 2021; Marler and Parry, 2015). In another definition, e-HRM is defined as targeting HRM practices that create consistent, efficient, high quality and long-term opportunities by using digital transformation and new technologies that develop together to meet organizational requirements and cover all HRM functions and content (Bondarouk and Brewster, 2016: 2659). If e-HRM research is to be realistic and closely related to the current order, we need a much broader focus on overcoming these assumptions. E-HRM has emerged as a planned outcome of strategic decisions regarding HRM services in organizations (Marler & Parry, 2016). The use of digital technologies in the HR field has resulted in the transformation from HRM to e-HRM or digital HRM, while at the same time benefiting from the full potential of technology development by providing greater flexibility (Hempel, 2004).

The use of digital technologies in the field of HR management is increasing and becoming more widespread. According to a global study by Bersin (2017), 74% of more than 7,000 companies in 130 countries around the world have drawn attention to the importance of incorporating digital technologies into their HR management practice. At the same time, as experts emphasize, HR today is not enough to just buy digital products, it has to learn to be “Digital” (Bondarouk, Parry, & Furtmueller, 2017). It is important for managers and leaders to first analyze their business and market situations, then decide on the right strategy and structure and involve their employees in the digital transformation. With the continuous change brought about by digital transformation, the increasing needs for knowledge, performance, productivity and skills cause great pressure on the workforce at all levels. Meeting these needs with digital learning platforms, organizing special trainings and development programs for employees and managers in accordance with their levels and branches can improve the internal training of enterprises. These new training methods support the acquisition of skills based on the strategies followed by organizations and which training systems cannot provide them (Gow & MacDonald, 2006). HR professionals need to familiarize the workforce with the need to develop skills and provide them with training and development interventions to provide what is best suited for the organization. E-learning, online learning, webinars, distance or online education and development programs, e-skills as well as certificates should be accepted by organizations (Charalampous et al., 2019).

2.1 Digital platform usage

The management practices of HR on digital platforms, which focus on managing both the management of the business and the people who will do the work, are carried out through the website or applications (Williams et al, 2021). HRM faces significant challenges in which digital platform companies apply HRM practices to attract or manage their own-account employees, and little empirical research has been done on this topic (Williams et al, 2021:4137). Digital platforms have developed as a workforce resource in traditional organizations to avoid the difficulties experienced in the implementation of HR practices such as employee compensation, performance management, recruitment, selection, training and job design (Ellmer and Reichel, 2018). Online platforms are used in

HRM activities such as personnel selection, recruitment and interview, communication, talent management, performance appraisal and wage management that organizations perform.

Due to the use and increasing importance of mobile devices, businesses have started to use mobile application platforms to manage their human resources activities, and Mobile Human Resources Management (mHRM) applications have developed rapidly due to their easy accessibility and cost-effectiveness (Shankar and Nigam, 2022). Aguinis and Lawal (2013) outlined a research agenda to explore how HR practices are applied to jobs accessed through digital platforms. They demand a better understanding of recruitment and selection and highlight how intermediary platforms can potentially confuse person-organization fit assessments. Other scholars have similarly questioned whether relevance can be assessed in the primitive and seemingly over-efficient recruitment processes of digital platforms (Duggan et al., 2020; Howcroft & Bergvall-Kåreborn, 2019).

With the development of mobile and cloud technologies, the first effect of digitalization on HR practices has manifested itself as “smart recruitment” in the recruitment process. Bersin (2017) stated that approximately 40% of companies use artificial intelligence in the recruitment process. Organizations such as IBM, SAP, Facebook, GE and Hilton around the world are known to use digital technologies to monitor, interview and recruit new talent. It is known that applications developed with artificial intelligence technologies are useful for the interview process by evaluating candidates' word choices, speech patterns and facial expressions in personnel selection. A short video interview with candidates turned facial movements and word choice into thousands of data points, contributing to the candidate evaluation process scientifically. In this way, it can be said that artificial intelligence helps the human resources department to make a selection by analyzing the qualifications required by the positions according to the success criteria, and then evaluating the existing talent pool to identify suitable candidates with sufficient skills and experience. It can also help gather data on behavioral aspects of potential candidates through machine learning and determine whether these candidates have a suitable culture for the organization. Artificial intelligence helps HR departments improve these processes effectively by redefining the process from pre-employment assessment and interviewing to recruitment and orientation training.

According to Mitrofanova, Konovalova and Mitrofanova (2019), the applications most used by human resources departments around the world, in the recruitment process; e-Personnel Recruiter, an online system that automates routine recruitment processes, is used. Chatbot services are chatbots that are designed to simulate human behavior while communicating with one or more interlocutors, where users communicate as if they are texting with a human being in a digital environment and are used for various purposes such as getting information on a subject and taking action. Firstjob Mya can eliminate 75% of the questions people ask in the recruitment process, Wade and Wendy chat bots chat with candidates and help them. As Candidate Tracking Systems, platforms such as Lever: Recruiting Software are used to manage the flow of candidates, while platforms such as Pymetrics and HireVue, which are candidate screening platforms containing various tools, are used to test candidates before the interview.

Solutions made available to HR are offered by large developers Microsoft (MSFT, NASDAQ), SAP (SAP, NYSE), IBM (IBM, NASDAQ), as well as relatively small developers Workday (WDAY, NYSE). The SAP Resume Matcher application from the developer of SAP SE (SAP, NYSE) can review information about job responsibilities and skills in a particular position, analyze data from thousands of resumes provided by the HR department, and rank candidates.

A product of Entelo Inc. is able to analyze publicly available information on 300 million potential candidates and select those that are appropriate for the company's requirements. Unlike classical learning systems, new and modern learning systems including Pathgather, Degreed, SAP Jam, Oracle's Video Learning, Workday Learning, Skillsoft's new learning platforms entered the market, providing educational materials, video and mobile learning solutions with Coursera, Udacity, EdX , Udemy, and

many other vendors have provided micro-learning and new solutions to integrate and use the growing library of web-based external Massive Open Online Course (MOOC), allowing employees to interact online with experts in the workspaces they need and at a time convenient for them.

Another important aspect is Robotic Process Automation, an intelligent artificial intelligence or machine learning technology that learns how a worker does a repetitive task. Robots are especially helpful in processes that require a lot of data processing. These robots, which can perform many predefined tasks in various environments, can check e-mails, connect with allowed applications, perform calculations and even gain the ability to issue invoices by verifying invoice amount. Robots can perform functions such as collecting data, running reports, copying data, checking data for completeness, reading, processing and sending emails, entering data into HR or payroll systems, and pre-filling forms. In addition, cloud technology-based solutions such as Microsoft 365 can help employees work and collaborate at the highest level without wasting time, creating a more streamlined work environment. AI tools can also help the HR function do personalized career development with different learning and development programs that will increase employee productivity and job satisfaction; reduce the level of work-related stress and employee turnover, and many other benefits. Progress in the use of digital technology is not an option or preference for businesses, it can be said that it is at the forefront of competition and sustainability strategies.

It can be said that there has been a revolution in the field of recruitment in recent years with the use of internet technologies, especially social media networks (SNW) such as Facebook, LinkedIn and Twitter. This has led to the emergence of the e-recruitment method, which means that all activities related to the recruitment process are carried out with the help of internet technologies, including finding, attracting and evaluating potential candidates. E-recruitment, digital human resource management databases of job information, job opportunities advertisements for career portals, social networking pages, corporate websites, intranets and other online resources and e-recruitment systems to submit applications, track, evaluate and manage candidates, abstract the database, and applications such as web browsers (Holm, 2012). It will be the most important competitive advantage for businesses to research what they need to implement and identify the best strategies to recruit and retain the future workforce. With the virtual interview becoming the new normal, businesses will try to attract the best and most talented employees by finding ways to differentiate themselves in the recruitment interview process.

Although digital HRM practices differ according to organizations, the results of comparative analyzes of HRM practices among countries (Tayeb, 2001; Parry, 2011; Som, 2012; Yoon & Chae, 2012; Tijdens et al., 2013) also differ according to countries (Berber et al., 2013). With the spread of e-HRM studies, a research area focused on cross-country comparison has emerged (Parry, 2011; Basu, 2019). The results of implementing e-HRM in different countries vary, and it is stated that e-HRM has gained importance in different sectors around the world (Basu, 2019). Parry (2011) argued that past research focused on a single country on e-HRM and cross-country research is important so that the results of past studies can be generalized to other countries. It has been emphasized that studies analyzing the differences or similarities of e-HRM applications at the cross-country level are few and limited in scope (Parry, 2011). Because the national context is more effective on organizational and institutional performance with the use of e-HRM practices (Marler et al., 2009). However, it stands out that the differences or similarities of e-HRM applications between countries should be examined further in future research.

3 Research method

This study, in the exploratory-descriptive research category, used systematic literature analysis to

analyze academic and scientific production and was assisted by the bibliometric method. Systematic reviews are considered the gold standard because systematic literature analysis includes a more detailed literature review and less author bias. A systematic literature search is a literature search conducted in a database (Web of Science) that contains the structure of your search history, which provides the evidence on which your research is based, by aggregating appropriate search terms related to the topic to be researched. It is worth noting that science itself has a cumulative structure that grows based on previous studies. For this reason, in order for the sources of a scientific study to be accessible, controllable and measurable, the complete transmission of the bibliographic components of the source used is a requirement of the scientificity of the cumulatively growing knowledge. Bibliometrics, on the other hand, is defined as a science with a quantitative methodology that focuses on the quantity of scientific publications with certain parameters for statistical purposes (Gauthier, 1998). Bibliometry is an innovative method often used for systematic literature reviews. For bibliometric analysis, Benckendorff and Zehrer (2013) refer to two basic bibliometric methods (evaluation and relational): evaluation techniques focus on the impact of academic studies by evaluating performance with productivity measures, impact measures, and mixed measures. Relational techniques, on the other hand, consider relationships among published research, taking into account citations, authors, author relationships, and keywords to conduct co-occurrence analyses. This study evaluates academic contributions in the field while revealing a retrospective view of publications based on electronic, digital or computer-based human resource management data. In this way, the relations between the publications are determined and the structure and evolution of the research field are investigated. As a result of extensive research, it was decided which keywords to be used. Keywords decided by the author are specified for obtaining the publications. The data source for the study is the Web of Science (WoS) database from 2012 to 2022. WoS is one of the most popular and reliable scientific citation index databases in the world (Fouroudi et al., 2020). Due to the comprehensiveness of our research themes, WoS database publications were sufficient for our research. Many of the researched keywords were determined to answer the research questions to fill the gap in the field.

The research statements used in this study are: “Electronic Human Resource Management Practices”, “e-HRM Practices”, “Computer-based Human Resource Management Practices”, “Virtual Human Resource Management Practices”, “Web-based Human Resource Management Practices”, “Digital Human Resource Management Practices”, “Digital Human Resource Management Application”, “Electronic Human Resource Management Application”, “Digital Human Resource Management Strategies”, “Web-based Human Resource Management Application”, “Digital Human Resource Management Implementation”, “Mobile Human Resource Management Application”. The inclusion criteria were refined by: DOCUMENT TYPES: (ARTICLE OR PROCEEDINGS PAPER OR REVIEW) AND LANGUAGES: (ENGLISH). Timespan: 2012-2022. Indexes: SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, BKCI-S, BKCI-SSH, ESCI, CCR-EXPANDED, IC. Three types of filters were used to select the documents to be reviewed. All duplicate documents are excluded. All headlines were read and articles unrelated to the theme were excluded. All abstracts and keywords were read, and articles unrelated to the theme were excluded. The research model is shown in Figure 1.

The research search and selection process was carried out by following the sequential processes in figure 1. Search terms are terms that are in the literature and have been studied. A large number of related topics have been identified within the scope of the research, and they have not been added to the list due to the lack of relevant publications in the database query. In the study, the number of publications found as a result of the database query for each research term is listed. VOSviewer, one of the visualization tools used in bibliometric mapping, was used. The total number of publications found 3888 articles were found in the WoS database. Publication selection and reduction research was conducted in accordance with the widely used systematic literature search methodology (see Keupp et al. 2012; Nabi et al. 2017; Rod, 2016). It was reduced to 79 articles as a result of the search for

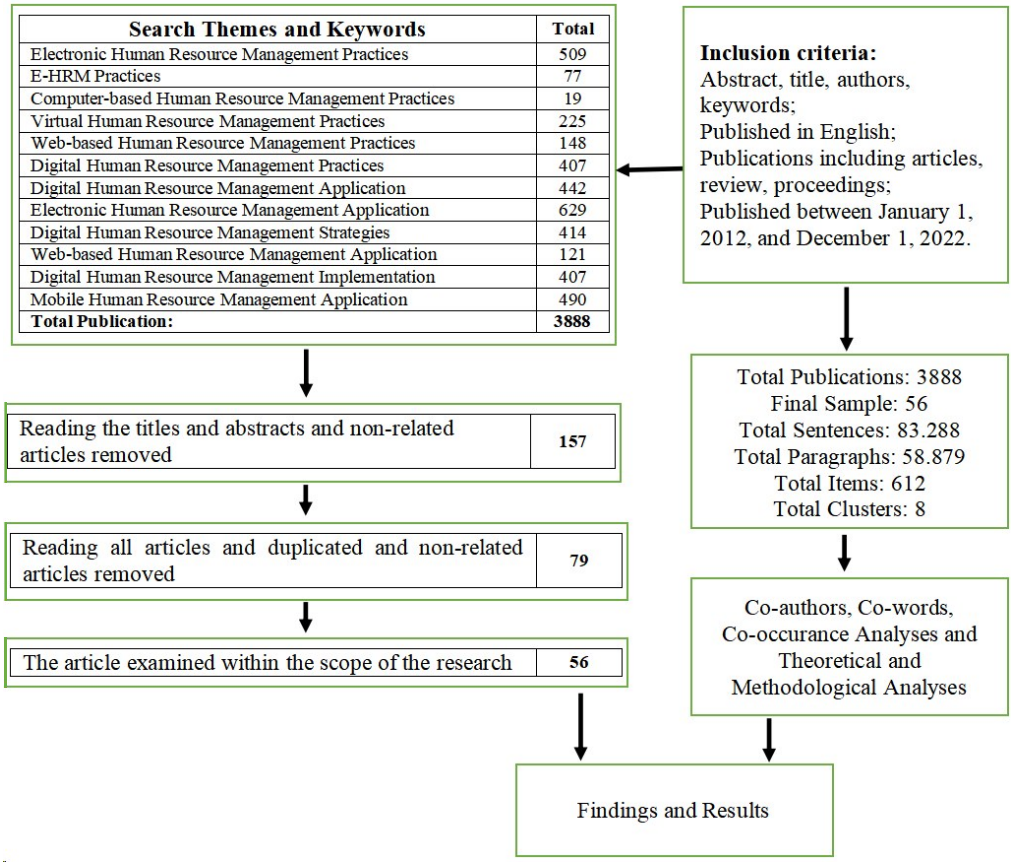

Figure 1 The process and research model of systematic literature review (own elaboration)

unrelated and duplicated publications. Finally, since the main focus of the research is to examine the subject within the framework of the management literature, which is considered worthy of examination within the scope of the research topic, each of the 56 publications, which is the final number of publications, was examined in detail by filtering the management and business categories from the articles obtained as a result of the Web of Science Core Collection search.

4 Findings

It is seen that the subject of e-HRM is the starting point in determining the research terms and the main starting point of the articles that guide the study, and the most cited publications in this field (table 1) are the studies on e-HRM. There are many definitions of e-HRM related and relevant to many fields, and the inclusive aspect of e-HRM as an "umbrella term" (Bondarouk and Ruël, 2009). The literature tries to explain the relationships between new technologies developing with digital transformation and HRM strategies, policies and practices with the new concept defined as e-HRM (Ruël et al. 2004:368). According to Fregnan et al. (2020) looks for new e-HRM research and applications, revealing the effects of strong business transformations in the twenty-first century on HRM and its reflections on HRM organizational life confronting the new technologies brought by the 4.0 Industrial Revolution. Although the umbrella term inspired the emergence of the work, the topics that focus on have much different content and scope. There are extensive and very useful studies in the literature to examine the definition, nature, content and configuration of e-HRM (Strohmeier, 2007; Bondarouk and Ruël, 2009;

Van Geffen et al. 2013; Marler and Fisher, 2013; Ruël and Bondarouk, 2014). It was concluded that most of the research on the use of technology to support HRM took place only in the last 15-20 years and was in response to the use of the web as a tool for the presentation of HR Information Systems (Bondarouk et al. 2017). In Table 1, the most cited publications (for the last 10 years (2012-2022)) in the WoS database records related to the main concept that forms the basis of the study are listed.

Table 1 The most frequently cited publications

No.	Source Year	Publication	Journal	Total Citation in WoS
1	Marler and Fisher (2013)	An evidence-based review of e-HRM and strategic human resource management	*Human Resource Management Review*	108
2	Parry and Tyson (2011)	Desired goals and actual outcomes of e-HRM	*Human Resource Management Journal*	97
3	Bondarouk, Parry, and Furtmueller (2017)	Electronic HRM: four decades of research on adoption and consequences	*The International Journal of Human Resource Management*	84
4	Parry (2011)	An examination of e-HRM as a means to increase the value of the HR function	*The International Journal of Human Resource Management*	72
5	Marler and Parry (2016)	Human resource management, strategic involvement and e-HRM technology	*The International Journal of Human Resource Management*	68
6	Stone and Dulebohn (2013)	Emerging issues in theory and research on electronic human resource management (eHRM)	*Human Resource Management Review*	64
7	Bondarouk and Brewster (2016)	Conceptualising the future of HRM and technology research	*The International Journal of Human Resource Management*	57
8	Bondarouk and Ruel (2013)	The strategic value of e-HRM: results from an exploratory study in a governmental organization	*The International Journal of Human Resource Management*	55
9	Lin (2011)	Electronic human resource management and organizational innovation: the roles of information technology and virtual organizational structure	*The International Journal of Human Resource Management*	54
10	Heikkila and Smale (2011)	The effects of 'language standardization' on the acceptance and use of e-HRM systems in foreign subsidiaries	*Journal of World Business*	41
11	Bondarouk, Harms and Lepak (2017)	Does e-HRM lead to better HRM service?	*The International Journal of Human Resource Management*	37
12	Schalk, Timmerman and van den Heuvel (2013)	How strategic considerations influence decision making on e-HRM applications	*Human Resource Management Review*	36
13	Ruel and van der Kaap (2012)	E-HRM Usage and Value Creation. Does a Facilitating Context Matter?	*German Journal of Human Resource Management: Zeitschrift für Personalforschung*	35
14	Obeidat (2016)	The link between e-HRM use and HRM effectiveness: an empirical study	*Personnel Review*	32

Table 1 The most frequently cited publications(Continued).

No.	Source Year	Publication	Journal	Total Citation in WoS
15	Dery et al. (2013)	Lost in translation? An actor-network approach to HRIS implementation	*Journal of Strategic Information Systems*	27
16	Russo et al. (2018)	Individual perceptions of HR practices, HRM strength and appropriateness of care: a meso, multilevel approach	*International Journal of Human Resource Management*	25
17	Wiblen (2016)	Framing the usefulness of eHRM in talent management: A case study of talent identification in a professional services firm	*Canadian Journal Of Administrative Sciences-Revue Canadienne Des Sciences De L Administration*	21
18	Wirtky et al. (2016)	On the Untapped Value of e-HRM: A Literature Review	*Communications of The Association For Information Systems*	21
19	Bissola and Imperatori (2014)	The unexpected side of relational e-HRM Developing trust in the HR department	*Employee Relations*	21
20	Strohmeier and Kabst (2014)	Configurations of e-HRM - an empirical exploration	*Employee Relations*	20

According to Table 1, the most cited work was done by Marler and Fisher (2013), published in the *Human Resource Management Review*, and received 108 citations. Since citation is accepted as the documentation of the information used, it is of primary importance for those who want to access scientific information. The fact that all of the cited journals are in the field of management and business, it is seen that they focus on the employees and the regulations in the workplace.

Table 2 The list of the most frequently seen keywords

Keywords	Occurrences	Total link strength
Management	304	1491
Performance	157	967
Impact	164	948
Human-resource management	107	691
Technology	114	679
Systems	98	567
Model	106	529
Innovation	79	500
Human resource management	101	460
Implementation	74	451
Challenges	70	431
Information	76	408
Information-technology	59	392
Framework	67	387
Knowledge	72	376
Work	62	355
Future	53	344
Care	63	305
Internet	53	303
E-hrm	48	273

Table 2 shows the first 20 terms and total link strengths with the most common occurrences in the whole data set obtained from 3888 articles evaluated for this study. The purpose of this analysis is to reveal around which expressions a very broad scope-oriented database query gains meaning. Identifying common formations plays an effective role in evaluating current and valid trends in the literature. At the same time, the network map of this co-occurrence is in Figure 2.

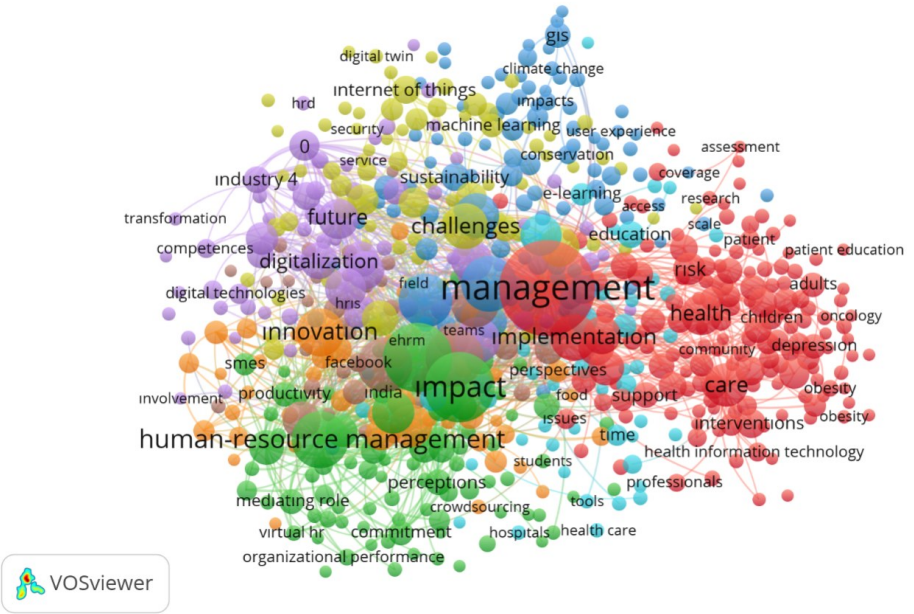

Figure 2 Co-occurrence network analysis of keywords

The common occurrence of 8 clusters related to the current electronic human resource management applications literature is visualized in Figure 2. It is aimed to establish the similarity between the two analysis units and to reveal the relational closeness of all similarity situations in the data set with the match-finding analysis. It is possible to say that co-word analysis was used in our study. It is possible to say that cointegration analysis, which is based on the principle of performing match analysis on keywords, is a method applied in many fields today (Daim et al., 2019). The method, which can be used for various purposes such as understanding the conceptual development of the field, and determining the relationships between concepts and working clusters in the field, is among the most preferred methods, especially for identifying research focuses. In the joint formation network image, it has been proven that the concepts in the field of information technologies and human resources management have gained prominence in the last ten years. The importance of technological concepts, which are important for all social and organizational life, especially today's human resources management practices, and whose benefits are increasing day by day, is clearly seen in Figure 2.

According to Figure 2, the bibliographic network map of the keywords is shown. A bibliometric map visualizes literary objects, object groups, and the relationships of objects with each other. A bibliometric map is a visual representation of bibliometric data. In order to create a comprehensive bibliometric map, it is necessary to show not only the objects but also the relations of the objects with each other. In total, 612 items and 8 cluster formations were observed. Accordingly, density is seen in the cluster with the most items (red). In the first cluster, management, implementation, health, risk, care expressions; in the second cluster (green) impact, human resource management, e-hrm, performance items; in the third cluster (blue) systems, framework, sustainability, science items; in the fourth cluster;

challenges, big data, artificial intelligence, Internet of Things, blockchain, cloud computing, digital twin, machine learning, digital transformation; in the fifth cluster, industry 4.0, digitalization, transformation, big data analytics; in the sixth cluster (light blue) culture, education, e-learning is innovation, information technology, information systems, entrepreneurship; in the seventh cluster (orange), and e-hrm, human resource management, facebook, talent management and web-based recruitment in the last cluster (magenta). Considering the distribution of expressions in the clusters, it was revealed that the fourth and fifth clusters were the clusters of technology and informatics concepts.

4.1 Theoretical studies and theories used in the studies reviewed

Old and well-established theories that are the most fundamental in technology and information-based HRM studies are critical contact theory, objective factor theory or objective theory of job choice, subjective factor theory, job choice theory (Behling et al., 1968), recruitment theory (Barber, 1998), signaling theory (Spence, 2002), but according to critical contact theory, job applicants make decisions about the desirability of a particular employer based on moments of interaction with the organization (Behling et al., 1968). Signaling theory suggests that information is important because job seekers faced with uncertainty and incomplete information use available information as signals about jobs and organizational qualifications (Rynes, 1991; Spence, 1973). During the signaling process, the sender decides what information is to be shared and how this information is transmitted (signal). Although this research was based on the last ten years, it was seen that the research and publications related to the field within the scope of research statements belong to the last few years. In this way, it has been a chance to reveal what the theoretical foundations of the most up-to-date research have been made in the last few years. The theoretical background of their work contributes to the more complex and superficial theory of HR practices in today's conditions. Different from the overview analyzes of Bondarouk and Brewster (2016) and Bondarouk and Furtmueller-Ettinger (2012), their e-HRM studies show that the research reveals contradictory institutional realities, while the adoption of e-HRM is beneficial in some cases (cost savings, efficiency, flexible services, employee involvement) can in some cases lead to negative consequences (job stress, more HRM management and frustrations with technology features). One of the biggest reasons for this is that e-HRM has not been seriously theorized enough, but most of the variation is likely due to the studies not paying enough attention to context.

Ruël, Bondarouk and Looise (2004: 368) defined e-HRM as 'a way of implementing HRM strategies, policies and practices in organizations with the informed and direct support and full use of web-based channels'. Overall, the new enhanced e-HRM will provide great insights into different types of information and can empower technology users to conduct different types of analysis on their HRM data. However, e-HRM may also impose extra responsibilities on its users regarding their own well-being, may require extra flexibility and will ultimately depend on their preferences to share information about HRM. E-HRM technologies are technologies developed for HRM policies, practices and rules, and adding a new technological dimension to HRM has made a challenging HRM context more complex and difficult. It has been stated that most of the research on e-HRM is conducted in a case study style (Bondarouk & Brewster, 2016). The diffusion of e-HRM was extensively studied in various international contexts (Florkowski & Olivas-Luján, 2006; Ruël & Bondarouk, 2012). Multinational companies are trying to implement e-HRM in an attempt to standardize their HRM practices among themselves.

Most of the articles in this group were written with multiple theoretical frameworks (Wang et al., 2022; Ni, 2022; Nankervis and Cameron, 2022; Konovalova et al., 2022; Ziebell et al., 2019; Qalhati et al. 2020; Vahdat, 2022). It is seen that technology and communication topics such as Internet of Things (IoT), cloud computing, big data, HR informatization, artificial intelligence, mobile communication network, and digital transformation have been intensified in the fields associated with digital HRM

applications since the last few years. However, it is seen that the studies that contribute to the theory in different ways, such as employee's performance (Qalhati et al. 2020), engagement, wellbeing, discrimination are associated with HRM strategy (Konovalova et al. 2022).

Table 3 Theoretical studies

Related Theoretical framework	Related Terms	Studies	Typology of study	Aim of the study	Main findings and propositions
Internet of Things, Internet, and mobile communication network	Human Resources Management Strategy	Wang et al. (2022)	Literature Review	Determine the impact of the IoT industry on HRM and its competitiveness in the market	Providing IoT theory support to enhance market competitiveness of the e-HRM strategies
Cloud computing	Human Resource Management Information System	Lv et al. (2018)	System Analysis	Create a HRM information system under the framework of cloud computing	HRM information system based on cloud computing can help enterprises to complete the task of human resource management efficiently.
Human resources informatization	Human Resource Management Platform	Liu et al. (2022)	Twin Network and Mobile Edge Comput-ing	Create an enterprise HRM platform application based on twin network and mobile edge computing	Using network communication technology to build a management information system significantly optimizes and improves the efficiency of enterprise human resource management
Human resources informatization	Human resource digital practices	Williams et al. (2021)	Qualita-tive, Com-parative Analysis	Implement HRM practices to attract or manage the quality of self-employed workers for Digital platform companies	Combined traditional recruitment practices with new forms of algorithmic shortlisting resulting in industry-specific risks for platform workers
Big data Digital HRM	Human resource management strategy	Ni (2022)	Deep Neural Network	Combine digital HRM knowledge to research human-job matching systematically	Intelligent digital means such as 5G communication, cloud computing, big data, neural network, and user portrait, this article proposes the design of the corresponding digital transformation strategy of HRM
Digital transformation and HRM	Human resource management strategy	Nankervis and Cameron (2022)	Mixed Methods	Identify the HR strategies and functions most likely to be affected by emerging digital technologies and explores the competencies and capabilities required by present and future HR professionals to transform these changing functions	The results show the patchy uptake of smart technologies, artificial intelligence, robotics and algorithms (STARA) technologies in the work-place, and the expected use of many of these technologies is more likely than it is now.
Employees Performance	HR Practices	Qalhati et al. (2020)	Literature review	Determining the impact of technology on HR functions	HR managers have the responsibility of maximizing the use of this technology in order to reap more benefits to the business community

Table 3 Theoretical studies(Continued).

Related Theoretical framework	Related Terms	Studies	Typology of study	Aim of the study	Main findings and propositions
Cloud Computing, Digital HR Transformation	e-HRM Implementation	Ziebell et al. (2019)	system-atic litera-ture review	(e-HRM) solutions are used to map a variety of HR processes.	Identified and reviewed articles under e-HRM, IT technology, and HR journals, it was possible to identify relevant controversial themes and gaps as well as limitations.
Human resources management, Cloud computing, Teleworking	HRM techniques	Vahdat, (2022)	systematic literature review	investigate some key consequences resulting from COVID-19's effect on multiple HRM roles, how technology is empowering and fostering HRM	Changes in the environment and the various challenges and requirements faced by businesses. The challenge of complying with social distancing and limiting face-to-face business services has been identified as the challenge of the COVID-19 pandemic period.
Artificial intelligence, Digital humanism, experience management, engagement, wellbeing, discrimination, HR management strategy	Human Resources Management Strategy	Konovalova et al. (2022)	Literature review	Large companies and even midsize organisations with employees spread across geographic regions are investing in cloud, mobile and AI to offer integrated HRM services in real-time and seamlessly	AI revolution makes HR processes less mechanical and more human-centred.

Numerous studies have been conducted showing that e-HRM applications differ by country (Nagendra and Deshpande, 2014; Bondarouk et al., 2016; Alam et al., 2016; Nura and Osman, 2013). In HRM strategies, the IoT industry is seen as one of the important areas in the future development of science and technology and the banking industry (Wang et al., 2022). According to Wang et al. (2022), the IoT industry uses the AHP method to analyze and evaluate HRM strategies, and combines the multi-objective genetic algorithm for IoT enterprise HRM strategy optimization. Cloud computing, big data management and processing are popular technologies in recent years and are widely used at the enterprise level. Cloud computing technology offers a solution that increases productivity by processing and storing big data in the field of human resources management (Lv et al., 2018). Ziebell et al. (2019) stated that there may be different strategic approaches for the adoption of e-HRM, and the Technology Acceptance Model (TAM) stands out due to its simplicity of use and many benefits. Virdyananto et al. (2017) examined the effect of using the TAM approach in combination with Technology Acceptance and Use Theory on e-HRM. TAM was used to analyze the contribution of e-HRM to business performance (Njoku, 2018). Business websites contribute to the clarity of first impressions for job applicants, and the adequacy of the features and content of these websites affect the attitudes and impressions of viewers towards the organization. Recruitment studies available in the literature focus on website features such as usability and aesthetics. It is aimed to implement HRM practices for digital platform companies to attract self-employed employees or to manage their quality, and combined traditional recruitment practices with new forms of recruitment resulting in industry-specific risks for platform employees (Williams et al., 2021).

From the theory development and adoption constraints, the regression analysis studied by Lin (2011) has a positive assessment that e-HRM is dependent on the adoption of both information technologies and a virtual organization. The use of artificial intelligence in HRM application areas has

become widespread in recent years. Especially in the recruitment and selection phase, it provides benefits and timesavings in identifying the best candidates. By using artificial intelligence, accurate results have emerged in the stages of training suggestions for the personnel and the most necessary trainings, checking and evaluating the candidates in the recruitment process. Konovalova et al. (2022) made a significant contribution to the literature in the selection of candidates in the recruitment process by using artificial intelligence technology, to prevent discrimination based on gender, age, race and ethnicity, to reduce human prejudice and to provide objective representation based on data. Because identifying, prioritizing and retaining high-potential employees by using artificial intelligence will increase productivity as well as reduce prejudices against bad impressions of recruitment processes (Konovalova et al., 2022).

4.2 Empirical qualitative studies

Empirical studies have benefited from multiple theoretical approaches investigating the impact of Human Resource Information System (HRIS) functions on current concepts. The four articles are qualitative empirical method studies and are based on interpretive case study, qualitative content analysis, and empirical case analysis. The main and common finding of imperial studies is to develop and contribute to the development of HRIS. The complexity of HRIS implementations, the inability to adequately manage user resistance according to metrics and an effective change management process are among the reasons why HRISs do not realize their expected potential (Dery et al., 2013). Case study is a useful method that provides convenience in revealing situations that cause delay and uncertainty in the development of HRISs, taking into account factors such as organizational strategy and structure, workforce size and composition.

Table 4 Qualitative studies

Related theoretical framework	Related terms	Studies	Typology of study	Aim of the study	Main findings and propositions
Actor-network theory (ANT) , HRIS, Social constructionist views	Digital human resources management Implementation	Dery et al. (2013)	Interpretative case study	The relationships between HRIS and HR function are examined by applying ANT to an HRIS implementation project.	ANT provides a better understanding of HRIS applications.
HRIS	Digital human resources management Implementation	Bhattacharyya et al. (2021)	Qualitative content analysis method	It is an article aimed at implementing an internet-based ICT tool known as the HRIS in 2017 to improve human resource management in the healthcare industry.	The implementation of a digital human resource management tool in the public health sector plays a vital role in the integration of ICT into health services. A well-defined monitoring strategy and inter-ministerial policy coordination to create a conducive environment are crucial for strengthened implementation.
Artificial intelligence, Industry 4.0, Employee experiences, Technostress, Human resources	e-HRM strategies	Malik at al. (2022)	Semi-structured interviews	Develop a practical understanding of the positive and negative employee experiences due to artificial intelligence (AI) adoption and the creation of technostress	Prominent adverse impacts of the adoption of AI like the potential risk of data security breaches, drastic organizational changes resulting from digital transformations and job risk and insecurity often trouble the employees.

Table 4 Qualitative studies (Continued)

Related theoretical framework	Related terms	Studies	Typology of study	Aim of the study	Main findings and propositions
Total quality management, Talent analytics in HR	HR Practices	Saputra et al. (2022)	a case study and experimental analyses	Help the HR department digitize and automate their analytical processes based on a big-data-analytics framework	Big data analytics is important in developing and advancing a company's other capabilities, especially its human resource management practices.

It is evident that qualitative empirical studies have a very broad scope of theories associated with HRM practices and strategies. Among them, Bhattacharyya et al. (2021) wrote an article aiming to implement an internet-based ICT tool known as the HRIS in 2017 to improve human resource management in the healthcare industry. According to this article, the implementation of a digital human resource management tool in the public health sector plays a vital role in the integration of ICT into health services. A well-defined monitoring strategy and inter-ministerial policy coordination to create a conducive environment are crucial for strengthened implementation (Bhattacharyya et al., 2021). According to Dery et al. (2013) focused on the management of users by associating the loss of functionality that may occur because of the automation of HRM processes with actor-network theory. Empirical research to date has also demonstrated the potential of an HRIS to enhance the strategic role of HR, highlighting the social construction of technology and the use of applied technologies, providing insights into the application and use of HRISs from a social constructivist perspective. However, it is more commonly used to increase the administrative and operational efficiency of HR processes. Malik et al. (2022) develop a practical understanding of the positive and negative employee experiences due to artificial intelligence (AI) adoption and the creation of technostress. We will see AI technologies very often in the field of HRM. AI is a field of informatics that focuses on solving problems, and an ever-evolving information technology that aims to think like humans and to perform tasks such as language processing, reasoning, problem solving, and learning.

4.3 Empirical quantitative studies

Recent discussions about technology and HRM strategies in businesses have emphasized the importance of social context in quantitative method studies, and it is seen that technologies have developed frameworks that accept both material and social characteristics, and studies conducted with the quantitative method are more than other methods.

Table 5 Quantitative studies

Related theoretical framework	Related terms	Studies	Typology of study	Aim of the study	Main findings and propositions
Human resource management, total quality management, and cognitive-motivational-relational (CMR) theory	Mobile human resource management application	Shankar (2022)	Quantitative research	Explore employees' perceived service quality towards mHRM app using the moderated-mediation framework.	The results show that navigation and communication significantly affect employees' mobile HRM usage behavior, and perceived values significantly mediate the relationship between perceived service quality and mHRM usage behavior.

Table 5 Quantitative Studies(Continued).

Related theoretical framework	Related terms	Studies	Typology of study	Aim of the study	Main findings and propositions
Resistance behaviour, Personal innovativness and Status quo bias (SQB) theory	Mobile HRM application	Shankar and Nigam (2022)	Quantitative research	Using the SQB theory, it examines the factors that influence HR professionals' resistance to mobile HRM implementation.	It showed that while regret avoidance, inertia, switching costs and perceived threat significantly affect HR professionals' resistance to adopt mobile HRM applications, they negatively affect the relationship between high personal innovativeness and resistance to mobile HRM application adoption.
Digital HR technology, HR transformation	Digital human resource technology	Jani et al. (2021)	Quantitative research	Investigate the impact of HR role and digital HR technology on successful HR transformation.	Digital human resource technology (HRT) can significantly improve the business results of India's Fortune 500 if mediated by different HR roles.
Digital transformation in relation to human resource management, Innovative work behavior	Human resource practices	Agustin et al. (2022)	Contingent approach	Some human resource practices in the digital age that favor the implementation of digital transformation.	Human resource practices partially mediate the relationship between strategic alignment and digital transformation.
HR theory, Organizational familiarity, Organizational image, Critical contact theory	e-HRM platforms e-HRM strategies	Intindola et al. (2019)	Quantitative research	Explore the effects of potential applicants' exposure to company information via different e-HRM platforms.	Dynamic e-HRM platforms would be influential in changing participants' organizational perceptions.
e-HRM	e-HRM practices	Imtiaz et al. (2019)	Quantitative research	Trace out the difference between the perceptions of university administrators from public and private sectors about the use of e-HRM practices.	The administrators from private sector universities have statistically significant differences over the administrators from public sector universities about the use of e-HRM practices.
Digital transformations, Employee involvement, Skills, Job design	e-HRM practices	Vereycken et al. (2021)	Quantitative research	Investigate the relationship between Industry 4.0 and employee engagement, job design and skills development HR practices.	Employee involvement shows a strong positive correlation with Industry 4.0, irrespective of the digital technology used, country or firm size. Secondly, weak but significant correlations are found with increasing job complexity and skill development.
HRM digitalization	e-HRM practices	Kucherov and Tsybova (2022)	Quantitative approach	Investigate the impact of e-recruitment practices on e-recruitment outcomes in Russian companies.	E-recruitment practices are positively related to e-recruitment outcomes.
Dynamic capability view, Sustainability	Human resource management (e-HRM) practices	Bag et al. (2022)	Partial least squares structural equation modelling method	Investigate whether electronic human resources management practices, opportunity-enhancing e-HRM practices and motivation-enhancing e-HRM can lead to the development of sustainable e-HRM systems.	Sustainable e-HRM systems can enhance firm performance.

Shankar (2022) conducted the first study on HRMS and service quality literature, investigating the service quality perception of employees towards mHRM usage behavior. It also expands the CMR theory literature by exploring the impact of service quality on usage behavior in the context of mobile HRM to create a new field. The study found results to help HR professionals manage their human resources effectively using the mHRM platform. According to the study of Shankar and Nigam (2022), in the transition from traditional HRM to mHRM applications of organizations where highly innovative employees are less resistant to mobile HRM application compared to low innovative employees, different understanding of innovative and non-innovative employees and increasing personal innovativeness of employees and need to organize seminars and training sessions on digital innovations has emerged.

Intindola et al. (2019), especially online recruiting and corporate websites, have gained intense interest and prevalence in the last decade. While previous recruitment efforts have seen users use company websites more and more for recruitment purposes over the past two decades, an increasing number of organizations have combined social media tools with company websites to integrate with newer forms of digital technology. Social media platforms like Facebook attract applicants, giving them many advantages in recruiting. HR transformation Jani et al. (2021), emphasizes the process of reorganizing the HR function in organizations for higher job performance. The HR transformation process emphasizes the different roles of HR. Jani et al. (2021) is a recent attempt to identify the relationship of HR transformation with the overall growth and development of organizations or firms. There is a positive relationship between talent enhancing e-HRM practices and sustainable e-HRM systems, and this relationship benefits organization.

5 Conclusion and future directions

Systematic bibliometric analysis aims to provide useful results by providing a practical review of the current literature and to guide future research directions by analyzing the current literature. Systematic bibliometric analysis, which have become increasingly important for scientific disciplines, enable researchers to access data belonging to a scientific field with a panoramic view. As the number of publications continues to increase, it becomes increasingly difficult to turn the data to be obtained from these publications into meaningful information with traditional methods. Determining the intellectual structure and the research front of scientific fields is not only for new research; it also gains importance for sectoral and public applications. The publications examined within the scope of the research are the current publications of recent years. Undoubtedly, this situation will cause this study to be a guide for future research. Adapting to the post-COVID-19 context can only be achieved if HR professionals ensure that the workforce is adaptable and agile during and after the crisis and at all stages of the crisis, the future of work, or the so-called "new normal", will result in networking, virtual interactions, cross-functional and centralized interactions. Therefore, to support this type of operation, HR professionals need to take advantage of this opportunity today and respond in a complex way to this pandemic or any crisis to create an effective and responsive organization that is well prepared to meet the new normal when a crisis arises.

It can be said that the digital transformation process is a time-consuming and long process, causing human resources practices to remain in the background. To avoid this situation, leaders and managers may need to attach importance to transparency and communicate frequently with all staff about the status of the process. The change strategy should be clear to everyone, not just the company's IT department. An effective communication plan can be a good way to ensure that employees are not excluded from this process. It can be said that employees have obligations to increase their knowledge and learn safe working principles in the digital field by improving themselves. Employees should actively participate in digitalization, not just as audiences. One of the ways to draw employees'

attention to digitalization is to make them a part of the production. Information technology human resources should certainly be included in digitization, but any employee can have a valuable impact on changing processes, especially if we are considering improving the digital workplace. The key to good technology is that it works for you. Regardless of the size of your business, it is important to start with a strong digital transformation strategy and choose the right technology to align with your business goals.

Thanks to the virtual and mobile platforms created by digital transformation, more competition awaits the workforce in the future. Digital transformation also has negative effects on the new working order. Employees in the digital age lack a sense of commitment to companies and therefore it is almost impossible to motivate them and expect them to contribute to the values, vision and mission of the companies. At the same time, they are not protected by trade unions or similar organizations and do not have common rights. Modern recruitment practices require competencies such as curiosity, flexibility, analytical perspective, innovation and openness to change. It will be more and more important for companies that employees in HR teams have these characteristics. In order for all HR processes to be carried out uninterruptedly and efficiently, user-friendly screens, HR software with a modular and interrelated structure, mobile application and complying with legal regulations should be preferred. On the other hand, the fact that various internal applications used by companies have an infrastructure that can be easily integrated and that they can provide tips to HR software will also increase efficiency and reportability. It is possible to say that the awareness of job seekers about benefiting from online resources will increase in the coming periods with the prominence of smart phone usage. We think that businesses will turn to move their growing HR platforms to cloud systems. This will bring flexibility in synchronizing and keeping data up-to-date. Every process of HR, without exception, is influenced by digital technologies to increase efficiency. However, in order to cope with a much more complex environment and competition, the comprehensive adoption of digital technologies must be supported by HRM practices while aligning with the organizational culture and core values of the company. Compliance of digital technologies with HR policies and practices should benefit the well-being of all employees and should be evaluated not only in terms of effectiveness and efficiency, but also in terms of perceived equity and fairness by all employees. Digital technologies affect the way employees respond to different stimuli and signals. At the same time, the spread of digital technologies affects the way people approach their work and there is a need for a workforce that is prone to new developments. This workforce will be referred to as the digital workforce. The increase in the use of digital technologies will affect the competencies and expectations of the digital workforce.

As a result, in the period we live in, fast learning, fast decision making and being more resistant have gained more importance. Undoubtedly, businesses that can adapt quickly to change will come out of this situation successfully. It can be said that it is not the time to stay behind, see what the others are doing, and decide accordingly, but to be one of the companies that make a difference by listening to the expectations of their employees and making the most appropriate decisions quickly. In addition, this study has some limitations. This study made an evaluation on the articles available in the literature. Therefore, the limitation of the obtained data set should be considered. However, only bibliometric analysis results are included in the study. For this reason, it may be recommended to combine bibliometric analysis with other types of analysis such as content analysis and meta-analysis in future studies.

About the Author

Hale Alan is an Associate Professor of Management and Strategy at Akdeniz University, where she is also a faculty member since 2018 and head of department since June 2022. She received a Ph.D. degree in management and organization at Başkent University in 2017 and her BA studies in human resource management at Gazi University in Turkey. E-mail: halealan@akdeniz.edu.tr

References

[1] Ageev, A.I., Averyanov, M., Evtushenko, S.N., & Kochetova, E.Y. (2017). Digital society: Architecture, principles, vision. Economic Strategies, 1: 114-125.

[2] Aguinis, H., & Lawal, S. O. (2013). ELancing: A review and research agenda for bridging the science–practice gap. Human Resource Management Review, 23(1): 6-17.

[3] Agustin, A.N. Jimenez, D.J. Fernandez, F. M. (2022). The role of human resource practices in the implementation of digital transformation. International Journal of Manpower, 43(2): 395-410.

[4] Alam, M. G. R., Masum, A. K. M., Beh, L. S., & Hong, C. S. (2016). Critical factors influencing the decision to adopt human resource information system (HRIS) in hospitals. PLoS One, 11(8): 1-22.

[5] Bag, S., Dhamija, P., Pretorius, J.H.C., Chowdhury, A.H. & Giannakis, M. (2022). Sustainable electronic human resource management systems and firm performance: An empirical study. International Journal of Manpower, 43(1): 32-51.

[6] Barber, A. E. (1998). Recruiting employees: Individual and organizational perspectives. SAGE Publications, Inc.

[7] Basu, M. (2019). Implementing e-hrm in cross-country environment: The key to organizational growth (January 5, 2019). Proceedings of 10th International Conference on Digital Strategies for Organizational Success.

[8] Benckendorff, P., & Zehrer, A. (2013). A network analysis of tourism research. Annals of Tourism Research, 43: 121-149.

[9] Behling, O., Labovitz, G., & Gainer, M. (1968). College recruiting: A theoretical base. Personnel Journal, 47:13-19.

[10] Berber, N., Đorđević, B., & Milanović, S. (2018). Electronic human resource management (e-hrm): A new concept for digital age. Strategic Management, 23(2): 22-32.

[11] Bersin, J. (2017). Digital HR: Platforms, people, and work. Global Human Capital Trends.

[12] Bhattacharyya, D.S., Dutta, G.K. (2021). Implementing a digital human resources management tool in the government health sector in Bangladesh: A policy content analysis, BMC Health Services Research, 21: 1346.

[13] Bissola, R., & Imperatori, B. (2014). The unexpected side of relational e-HRM. Employee Relations, 36(4): 376-397.

[14] Bondarouk, T. V., & Ruël, H. J. M. (2009). Electronic human resource management: Challenges in the digital era. International Journal of Human Resource Management, 20(3): 505-514.

[15] Bondarouk, T., Parry, E. and Furtmueller, E. (2017). Electronic HRM: Four decades of research on adoption and consequences. The International Journal of Human Resource Management, 28(1): 98-131.

[16] Bondarouk, T., Harms, R., & Lepak, D. (2015). Does e-HRM lead to better HRM service? The International Journal of Human Resource Management, 1-31.

[17] Bondarouk, T., and Brewster, C. (2016). Conceptualising the future of HRM and technology research. The International Journal of Human Resource Management, 27: 2652-2671.

[18] Bondarouk, T., & Furtmueller-Ettinger, E. (2012). Electronic human resource management: Four decades of empirical evidence. Best Paper Proceedings of the Academy of Management Meeting 2012, Boston, MA.

[19] Bondarouk, T., & Ruel, H. J. M. (2013). The strategic value of e-HRM: Results from an exploratory study in a governmental organization. The International Journal of Human Resource Management, 24(2): 391-414.

[20] Caligiuri, P., De Cieri, H., Minbaeva, D. et al. (2020). International HRM insights for navigating the COVID-19 pandemic: Implications for future research and practice. Journal of International Business Studies, 51: 697-713.

[21] Cedar C. (2006). Cedar crestone 2006 workforce technologies and service delivery approaches survey. Cedar Crestone.

[22] Charalampous, M., Grant, C. A., Tramontano, C., & Michailidis, E. (2019). Systematically reviewing remote e-workers' well-being at work: A multidimensional approach. European Journal of Work and Organizational Psychology, 28(1): 51-73.

[23] Daim, T., Bukhari, E., Bakry, D., VanHuis, J., Saadatmand, M., Yalcin, H., & Wang, X. (2019). Forecasting technology trends of the gap between science and technology: The case of software as a service e-commerce. Paper Presented at the Proceedings of the International Annual Conference of the American Society for Engineering Management.

[24] Dery, K., Hall, R., Wailes, N., & Wiblen, S. (2013). Lost in translation? An actor-network approach to HRIS implementation. Journal of Strategic Information Systems, 22(3): 225-237.

[25] Duggan, J., Sherman, U., Carbery, R., & McDonnell, A. (2020). Algorithmic management and app-work in the gig

economy: A research agenda for employment relations and HRM. Human Resource Management Journal, 30(1): 114-132.

[26] Ellmer, M., & Reichel, A. (2018). Mind the channel! An affordance perspective on how digital voice channels encourage or discourage employee voice. Human Resource Management Journal, 31(1): 259-276.

[27] Florkowski, G. W., & Olivas-Luján, M. R. (2006). The diffusion of human-resource informationtechnology innovations in non-US firms and US. Personnel Review, 35: 684-710.

[28] Fregnan, E., Ivaldi S., & Scaratti, G. (2020). Hrm 4.0 And new managerial competences profile: The comau case. Front. Psychol, 11: 578251.

[29] Foroudi, P., Reza, M., Mohammad, M. F., Sayabek, Z., & Solongo, M. (2020). Against the odds: Consequences of social media in b2b and b2c. In Beyond Multi-Channel Marketing. Bingley: Emerald Publishing Limited.

[30] Gauthier, E. (1998). Bibliometric analysis of scientific and technological research: A user's guide to the methodology. Science and Technology Redesign Project Statistics Canada.

[31] Grant, C. A., Tramontano, C., & Michailidis, E. (2019). Systematically reviewing remote e-workers' well-being at work: A multidimensional approach. European Journal of Work and Organizational Psychology, 28(1): 51-73.

[32] Gow, K., & McDonald, P. (2006). Attributes required of graduates for the future workplace. Journal of Vocational Education & Training, 52: 373-396.

[33] Heikkil€a, J. P., & Smale, A. (2011). The effects of 'language standardization' on the acceptance and use of e-HRM systems in foreign subsidiaries. Journal of World Business, 46(3): 305-313.

[34] Hempel, P. S. (2004). Preparing the HR profession for technology and information work. Human Resource Management, 43(2/3): 163-177.

[35] Holm, B. A. (2012). Towards an ubiquitous recruitment process and candidate relationship management. German Journal of Research in Human Resource Management, 26(3).

[36] Howcroft, D. & Bergvall-Kåreborn, B. (2019). A typology of crowdwork platforms work employ. Soc., 33(1): 21-38.

[37] Imtiaz, H., Iftikhar, A. B., Namra, M., & Ejaz, A. M. (2019). The comparison of the perception of university administrators about the use of e-human resource management practices. Journal of Organizational Behavior Research, 4(2): 98-105.

[38] Intindola, M. L., Lewis, G., Flinchbaugh, C., & Rogers, S. E. (2019). Web-based recruiting's impact on organizational image and familiarity: Too much of a good thing?. The International Journal of Human Resource Management, 30 (19): 2732-2753.

[39] Jani, A., Muduli, A., & Kishore, K. (2021). Human resource transformation in India: examining the role digital human resource technology and human resource role. International Journal of Organizational Analysis.

[40] Keupp, M.M., Palmie, M., & Gassmann, O. (2012). The strategic management of innovation: A systematic review and paths for future research. International Journal of Management Reviews, 14: 367-390.

[41] Konovalova, V., Mitrofanova, E., Mitrofanova, A., & Gevorgyan, R. (2022). The impact of artificial intelligence on human resources management strategy: Opportunities for the humanisation and risks. WISDOM-Special Issue 1(2): 88-98.

[42] Kucherov, D., & Tsybova, V. (2022). The contribution of e-recruitment practices to e-recruitment outcomes in Russian companies. Measuring Business Excellence, 26(3): 366-377.

[43] Lin, L. H. (2011). Electronic human resource management and organizational innovation: The roles of information technology and virtual organizational structure. International Journal of Human Resource Management, 22(2): 235-257.

[44] Liu, L., Sun, B, & Xu, Q. (2022). Mobile edge computing application in enterprise human resource management platform based on task scheduling algorithm. Mobile Information Systems, 22.

[45] Lv, Z., Tan, Z., Wang, Q., & Yang, Y. (2018). Cloud computing management platform of human resource based on mobile communication technology. Wireless Pers Commun, 102: 1293-1306.

[46] Malik, N., Tripathi, S. N., Kar, A. P., & Gupta, S. (2022). Impact of artificial intelligence on employees working in industry 4.0 led organizations. International Journal of Manpower, 43(2): 334-354.

[47] Marler J., Parry E., & Lepak D. (2009). E-HRM in different national contexts: Strategic trigger or strategic choice? Paper Presented at the Academy of Management Annual Conference, Chicago, USA.

[48] Marler, J. H., & Fisher, S. L. (2013). An evidence-based review of e-HRM and strategic human resource management.

Human Resource Management Review, 23(1): 18-36.

[49] Marler, J. H., & Parry, E. (2016). Human resource management, strategic involvement and e-HRM technology. International Journal of Human Resource Management, 27(19): 2233-2253.

[50] Mitrofanova, E.A., Konovalova, V.G., & Mitrofanova, A.E. (2019). Opportunities, problems and limitations of digital transformation of HR management. The European Proceedings of Social & Behavioural Sciences, 57: 1717-1727.

[51] Nabi, G., Linan, F., Fayolle, A., Krueger, N. & Walmsley, ´A. (2017). The impact of entrepreneurship education in higher education: A systematic review and research agenda. Academy of Management Learning & Education, 16: 277-299.

[52] Nagendra, A., & Deshpande, M. (2014). Human resource information systems (HRIS) in HR planning and development in mid to large sized organizations. Procedia: Social and Behavioral Sciences, 133: 61-67.

[53] Nankervis, A.R., & Cameron, R. (2022). Capabilities and competencies for digitised human resource management: Perspectives from Australian HR professionals. Asia Pacific Journal of Human Resources, 61(1): 232-251.

[54] Ni, Q. (2022). Deep neural network model construction for digital human resource management with human-job matching. Computational Intelligence and Neuroscience, 22.

[55] Nura, A. A., & Osman, N. H. (2013). Gauging the effect of performance management and technology based human resource management on employee retention: The perspective of academics in higher educational institutions in Sokoto State Nigeria. Asian Social Science, 9: 295-304.

[56] Obeidat, S. M. (2016). The link between e-HRM use and HRM effectiveness: an empirical study. Personnel Review, 45(6): 1281-1301.

[57] Parry, E., & Tyson, S. (2011). Desired goals and actual outcomes of e-HRM. Human Resource Management Journal, 21(3): 335-354.

[58] Parry, E. (2011). An examination of e-HRM as a means to increase the value of the HR function. The International Journal of Human Resource Management, 22(5): 1146-1162.

[59] Rod, I. (2016). Disentangling the family firm's innovation process: A systematic review. Journal of Family Business Strategy, 7: 185-201.

[60] Ruël, H., & van der Kaap, H. (2012). E-hrm usage and value creation. Does a facilitating context matter?. German Journal of Human Resource Management, 26(3): 260-281.

[61] Ruël, H., Bondarouk, T., & Looise, J. K. (2004). E-HRM: Innovation or irritation. An explorative empirical study in five large companies on web-based HRM. Manag. Rev., 15: 364-380.

[62] Ruël, H. J. M., & Bondarouk, T. (2012). A cross-national perspective on the intersection between information technology and HRM. In C. Brewster & M. Mayrhofer (Eds.), a handbook of research on comparative human resource management. Cheltenham: Edward Elgar, 416-448.

[63] Ruël, H., & T. Bondarouk. (2014). E-HRM research and practice: Facing the challenges ahead. In handbook of strategic e-business management. Springer Berlin Heidelberg, 633-653.

[64] Russo, S. D., Mascia, D., & Morandi, F. (2018). Individual perceptions of HR practices, HRM strength and appropriateness of care: Ameso, multilevel approach. The International Journal of Human Resource Management, 29(2).

[65] Rynes, S.L., (1991). Recruitment, Job Choice, and Post-Hire Consequences: A call for new research directions, in handbook of industrial and organizational psychology, M. D. Dunnette ve L. M. Hough, Eds., Palo Alto, CA: Consulting Psychologists Press.Rynes, S.L.; Miller, H.E.,recruiter and job influences on candidates for employment. Journal of Applied Psychology, 68(1): 147-154.

[66] Saputra, A., Wang, G., Zhang, J. Z., & Behl, A. (2022). The framework of talent analytics using big data. The TQM Journal, 34(1): 178-198.

[67] Schalk, R., Timmerman, V., & den Heuvel, S. van. (2013). How strategic consideration sinfluence decision making on e-HRM applications. Human Resource Management Review, 23(1): 84-92.

[68] Shankar, A. (2022). Does employees' perception towards mobile human resource management application quality affect usage intention? A moderated-mediation analysis. The TQM Journal, 34(1): 145-159.

[69] Shankar, A. and Nigam, A. (2022). Explaining resistance intention towards mobile HRM application: The dark side of technology adoption. International Journal of Manpower, 43(1): 206-225.

[70] Sharapova, N. V., Sharapova, V. M., & Shvetsova, D. M. (2017). Digital recruitment technologies. Creative Economy,

11(9): 953-966.

[71] Som, A. (2012). Organizational response through innovative HRM and redesign: A comparative study from France and India. The International Journal of Human Resource Management, 23(5): 952-976.

[72] Spence, M. (1973). Job market signaling. Quarterly Journal of Economics, 87: 355-373.

[73] Spence, M. (2002). Signaling in retrospect and the informational structure of markets. American Economic Review, 92 (3): 434-459.

[74] Stone, D. L., & Dulebohn, J. H. (2013). Emerging issues in theory and research on electronic human resource management (e-HRM). Human Resource Management Review, 23: 1-5.

[75] Strohmeier, S. (2007). Research in e-HRM: Review and implications. Human Resource Management Review, 17(1): 19-37.

[76] Strohmeier, E., & Parry, S. (2014). HRM in the digital age—digital changes and challenges of the HR profession. Empl. Relat, 6: 456-461.

[77] Strohmeier, S., & Kabst, R. (2009). Organisational adoption of e-HRM in Europe: an empirical exploration of major adoption factors. Journal of Managerial Psychology, 24(6): 482-501.

[78] Tijdens, G. K., De Ruijter, E., & De Ruijter, J. (2013). Comparing tasks of 160 occupations across eight European countries. Employee Relations, 36(2): 110-127.

[79] Qalhati, N., Karim, A. M., Al Mughairi, B., Al Hilali, K., & Hossain, M. I. (2020). Technology and HR practices in educational sector in sharqiya governate of oman. International Journal of Academic Research in Business and Social Sciences. 10(10): 435-443.

[80] Vereycken, Y., Ramioul, M., Desiere, S., & Bal, M. (2021). Human resource practices accompanying industry 4.0 in European manufacturing industry. Journal of Manufacturing Technology Management, 32(5): 1016-1036.

[81] Wang, J. Bai, W., & Liu, Y. (2022). Optimization for the human resources management strategy of the iot industry based on AHP. Computational Intelligence and Neuroscience.

[82] Wiblen, S. (2016). Framing the usefulness of eHRM in talent management: A case study of talent identification in a professional services firm. Canadian Journal of Administrative Sciences/Revue Canadienne des Sciences de l'Administration, 33(2): 95-107.

[83] Williams, P., McDonald, P., & Mayes, R. (2021) Recruitment in the gig economy: attraction and selection on digital platforms. The International Journal of Human Resource Management, 32(19): 4136-4162.

[84] Wirtky, T., Laumer, S., Eckhardt, A., & Weitzel, T. (2016). On the untapped value of it in HRM—A literature review communications of the association for information systems. CAIS, 38(2).

[85] Vahdat, S. (2022). The role of IT-based technologies on the management of human resources in the COVID-19 era. Kybernetes: The International Journal of Systems & Cybernetics, 51(6): 2065-2088.

[86] Van Geffen, C., Ruël, H., & Bondarouk, T. (2013). E-HRM in MNCs: What can be learned from a review of the IS literature? European Journal of International Management, 7(4): 373.

[87] Virdyananto, A. et al. (2017). User acceptance of human resource information system: An integration model of unified theory of acceptance and use of technology (UTAUT), task technology fit (TTF), and symbolic adoption. 2016 International Conference on Information Technology Systems and Innovation, ICITSI 2016-Proceedings, 1-6.

[88] Yoon, S. J., & Chae, Y. J. (2012). Management of paradox: A comparative study of managerial practices in Korean and Japanese firms. The International Journal of Human Resource Management, 23(17): 3501-3521.

[89] Ziebell, R. C., Albors-Garrigos, J., Schoeneberg, K. P., & Marin, M. R. P. (2019). Adoption and success of e-HRM in a cloud-computing environment: A field study. International Journal of Cloud Applications and Computing, 9(2): 1-27.